Methodology Document

SAAM Project 2026 — Part I: Standard Portfolio Allocation

Region: EUR | CO2 Scope: 1 + 2

1. Configuration Parameters

The following parameters govern the entire pipeline. Each represents a design choice that affects the investment set, estimation quality, and portfolio characteristics.

Parameter	Value	Rationale	Source
REGION	EUR	Group assignment	Project instructions
START_YEAR	2013	First allocation decision (Dec 2013 for 2014)	Project, Section 2.1
END_YEAR	2024	Last allocation decision (Dec 2024 for 2025)	Project, Section 2.2
ESTIM_YEARS	10	Rolling window = 120 monthly obs	Project, Section 2.1
MIN_OBS	36	Minimum 3 years of valid returns	Project, Section 2.1
STALE_THR	0.30	Max fraction of zero-return months (stricter than suggested 0.50)	Project, Section 1
LOW_FLOOR	0.50	RI values below this treated as NaN	Project, Section 1
LW_SHRINK_FLOOR	0.01	Minimum shrinkage intensity for Ledoit-Wolf	Ledoit & Wolf (2004)

2. Data Cleaning

2.1 Low Price Filter

All Total Return Index (RI) values below 0.50 are set to NaN. This prevents extreme or infinite returns from near-zero prices caused by rounding artefacts in Datastream.

Aspect	Advantage	Disadvantage
Return stability	Eliminates division-by-small-number artefacts that produce spurious 1000%+ returns	May exclude legitimate penny stocks from the universe
Consistency	Threshold is explicit and rule-based, avoiding ad hoc judgment per firm	The 0.50 cutoff is arbitrary; a lower value (e.g., 0.10) would retain more firms

Source: Project instructions, Section 1 ("I suggest treating all prices below 0.5 as missing values").

2.2 Forward-Fill Internal Missing Prices

For each firm, missing RI values that fall between the first and last valid observation are forward-filled from the most recent available price. This avoids look-ahead bias since we never use future prices to impute past gaps. Leading NaN (firm not yet listed) and trailing NaN (firm delisted) are not filled.

Aspect	Advantage	Disadvantage
Bias avoidance	Forward-fill uses only past information, preventing look-ahead bias	If a firm was suspended for months, this silently assumes zero return throughout the gap
Data coverage	Maximises the number of usable return observations per firm	Artificial zero-return stretches can deflate estimated variance, though this is mitigated by the stale price filter

Source: Project instructions, Section 1 ("I suggest using the number from the previous year"); Lecture 5, slide 19.

2.3 Delisting Detection and -100% Returns

Firms whose last valid RI falls before the end of the sample are flagged as potential delistings. At the first month after the last valid price, a -100% return is applied. All subsequent returns are set to NaN, effectively removing the firm from the portfolio.

Additionally, delistings are detected during the out-of-sample (OOS) year via the **fill_oos_returns()** function. For each eligible firm in the OOS period, if the return is missing but the previous month had a valid RI price, this is treated as a mid-year delisting with a -100% return. If the previous month also had no valid price (leading NaN), the return defaults to 0%.

Aspect	Advantage	Disadvantage
Survivorship bias	Captures full delisting loss instead of silently dropping the firm, correcting upward bias in returns	May misclassify temporary data gaps as delistings (e.g., a reporting delay)
OOS coverage	Extends delisting detection into the out-of-sample year, not just the estimation window	Conservative: any disappearing price triggers -100%, even if the firm merges at a premium

Source: Project instructions, Section 1 ("you should acknowledge that the price goes to 0 at the time of delisting, implying a realised return of -100%"); Shumway, T. (1997), "The Delisting Bias in CRSP Data", Journal of Finance, 52(1), 327-340.

2.4 Stale Price Filter

For each firm, the fraction of months with zero return ($|R| < 1e-10$) over the 10-year estimation window is computed. As per the teacher's instructions, firms exceeding 30% zero-return months are excluded from the investment set

Aspect	Advantage	Disadvantage
Illiquidity protection	More aggressive filtering (30% vs 50%) prevents the optimizer from loading on illiquid stocks with artificially low variance	Smaller investment universe, which may reduce diversification
Robustness	The threshold is computed on past data only (no look-ahead bias)	Sensitivity to the exact cutoff: 30% vs 50% could meaningfully change the eligible set

Source: Lesmond, D., Ogden, J. & Trzcinka, C. (1999), "A New Estimate of Transaction Costs", Review of Financial Studies, 12(5), 1113-1141.

2.5 Carbon and Revenue Data: Forward-Fill

Annual CO2 emissions (Scope 1 and 2) and revenues are forward-filled along the time axis. This means if a year's value is missing but prior years are available, the most recent value is carried forward. Leading NaN are not filled: firms without initial carbon data remain excluded until data becomes available.

Aspect	Advantage	Disadvantage
Data availability	Maximises the set of firms with carbon data, especially in earlier years when coverage is sparse	Stale carbon data may not reflect actual emissions trends (e.g., a firm that doubled emissions)
Revenue staleness	Revenue can be volatile year-to-year; carrying a stale number for 2+ years distorts carbon intensity	No practical alternative without look-ahead bias

Source: Project instructions, Section 1 ("When the missing value is between two available years or at the end of the sample, just use the number from the previous year").

3. Investment Set Construction

At the end of each year Y, the investment set for year Y+1 is determined by applying four sequential filters to all EUR firms in the static file. All filters use information available up to and including December of year Y, ensuring no look-ahead bias.

3.1 Filter Criteria

Filter	Description
Region	Firm belongs to EUR region (per Static_2025.xlsx assignment)
Valid price at Dec Y	RI at end of year Y is non-NaN (after low-price and forward-fill cleaning)
Sufficient returns	At least 36 valid monthly return observations in the 120-month estimation window ending Dec Y
Stale price	Fraction of zero-return months in estimation window does not exceed 30%
Carbon data	Both Scope 1 and Scope 2 CO2 emissions available at end of year Y (after forward-fill)

3.2 Carbon Data Requirement

The project instructs that the same investment set be used for both Part I (standard allocation) and Part II (carbon-constrained allocation). Therefore, even in Part I, firms without available carbon data at the end of year Y are excluded. This ensures portfolio comparability across parts.

Source: Project instructions, Section 2.1 ("because we want to use the same investment set for both parts of the project, exclude firms without carbon data available at the end of year Y").

4. Return and Covariance Estimation

4.1 Expected Returns

Expected returns are estimated as the per-firm sample mean of monthly simple returns over the 10-year rolling window ending at December of year Y. Each firm's mean is computed only over its available (non-NaN) observations. This is the ML estimator consistent with the project formula.

Source: Project instructions, Section 2.1; Lecture 5, slides 19 and 33.

4.2 Pairwise-Complete Covariance Matrix (Step 1)

The covariance matrix is constructed in two stages. The first stage follows the lecture method using pairwise complete observations. For each pair of firms (i, j), the correlation is computed over the

months where both firms have valid returns. Each firm's variance is computed over its own full available sample. The covariance is then reconstructed as Cov(i,j) = Corr(i,j) * sqrt(Var(i) * Var(j)).

This approach uses the ML estimator (dividing by n, not n-1), consistent with the project formula. Variances are computed over each firm's entire available sample, while correlations use the common sub-sample. This maximises data usage.

Aspect	Advantage	Disadvantage
Data efficiency	Uses all available observations per firm for variance, maximising estimation accuracy	The resulting matrix is NOT guaranteed to be positive semi-definite (PSD)
Pairwise correlation	Does not require all firms to have identical observation periods	Pairs with very few common observations produce noisy correlation estimates
ML estimator	Consistent with the project formula (1/T divisor)	Slightly biased downward in finite samples vs unbiased estimator (1/(T-1))

Source: Lecture 5, slides 20-21; Stambaugh, R.F. (1997), "Analyzing Investments Whose Histories Differ in Length", Journal of Financial Economics, 45(3), 285-331.

4.3 Ledoit-Wolf Shrinkage to Constant-Correlation Target (Step 2)

The pairwise-complete covariance matrix from Step 1 suffers from two problems: (a) it may not be PSD, making the optimization problem non-convex, and (b) it contains estimation error that the optimizer amplifies by overweighting assets with underestimated variance/covariance.

To address both problems, we apply Ledoit-Wolf shrinkage. The shrinkage estimator blends the sample covariance S with a structured target F:

Sigma_shrunk = delta * F + (1 - delta) * S

where F is the constant-correlation target matrix (each firm keeps its own variance, but all pairwise correlations are set to the average off-diagonal correlation), and delta is the shrinkage intensity estimated analytically.

The shrinkage intensity delta is computed using the Oracle Approximating Shrinkage (OAS) formula, which minimises expected squared Frobenius loss. The effective sample size T_eff is set to the median pairwise observation count, accounting for the unbalanced panel. A floor of LW_SHRINK_FLOOR = 0.01 prevents zero shrinkage.

As a final safety net, spectral clipping is applied: any remaining negative eigenvalues are set to a small positive floor (1e-10), guaranteeing PSD.

Aspect	Advantage	Disadvantage
PSD guarantee	Shrinkage + spectral clipping ensure the covariance matrix is always PSD, making the QP well-defined	Shrinkage introduces bias toward the structured target

Estimation error	Systematically reduces the extreme coefficients that mislead the optimizer	The constant-correlation target assumes all correlations are equal, which is a simplification
Analytical delta	No cross-validation or tuning needed; the optimal intensity is derived in closed form	OAS is an approximation; the exact Ledoit-Wolf formula requires complete data
Removes zero-substitution	The old approach of zeroing covariances for thin-overlap pairs asserted zero correlation; shrinkage instead pulls these toward a plausible average	Pairs with very few common observations still contribute noise, now partially dampened rather than eliminated

Source: Ledoit, O. & Wolf, M. (2004), "A well-conditioned estimator for large-dimensional covariance matrices", Journal of Multivariate Analysis, 88(2), 365-411; Ledoit, O. & Wolf, M. (2003), "Honey, I Shrunk the Sample Covariance Matrix", Journal of Portfolio Management, 30(4), 110-119; Chen, Y., Wiesel, A., Eldar, Y. & Hero, A. (2010), "Shrinkage Algorithms for MMSE Covariance Estimation", IEEE Trans. Signal Processing, 58(10), 5016-5029.

5. Minimum-Variance Portfolio Optimisation

5.1 Optimisation Problem

At the end of each year Y, we solve the following constrained quadratic program:

min alpha' * Sigma_Y * alpha subject to alpha' * e = 1, alpha_i >= 0 for all i

This is the long-only global minimum-variance (GMV) portfolio. The positivity constraint prevents short selling, which is required for institutional interpretability and for meaningful carbon footprint computation in Part II.

Source: Project instructions, Section 2.2; Lecture 5, slides 24-25, 34.

5.2 Solver Configuration

Setting	Value / Rationale
Method	SLSQP (Sequential Least Squares Quadratic Programming) via scipy.optimize.minimize
Jacobian	Analytical gradient: 2 * Sigma @ w (avoids finite-difference approximation)
Bounds	[0, None] for all weights (long-only, no upper bound)
Equality constraint	sum(w) = 1 (full investment)
Tolerance	ftol = 1e-10 (tight convergence)
Max iterations	1000
Initial guess	Drifted weights from previous year (warm start), or 1/N for first year

5.3 Warm Start from Drifted Weights

Instead of always starting from equal weights (1/N), the optimizer is initialised with the portfolio's actual weight composition at the end of the previous year (after 12 months of price-driven weight drift). This reflects the investor's current holdings before rebalancing.

Aspect	Advantage	Disadvantage
Convergence speed	Starting closer to the solution reduces iterations and avoids local-minimum traps in the non-convex landscape	If the drifted portfolio is far from optimal (e.g., due to a structural break), the warm start may bias the optimizer
Practical relevance	Mirrors real portfolio management: the rebalancing starts from where the portfolio actually is	Introduces path dependence into the optimisation sequence

Source: Standard practice in rolling portfolio optimisation; Jagannathan, R. & Ma, T. (2003), "Risk Reduction in Large Portfolios: Why Imposing the Wrong Constraints Helps", Journal of Finance, 58(4), 1651-1684.

5.4 No Upper Bound on Individual Weights

The optimisation imposes no maximum weight per firm. In principle, a single stock could receive 50%+ of the portfolio. This is technically correct per the project instructions, which only require non-negativity. However, it creates concentration risk.

Flag: In practice, the long-only constraint combined with Ledoit-Wolf shrinkage partially mitigates extreme concentration. Jagannathan & Ma (2003) show that the non-negativity constraint acts as an implicit form of shrinkage, improving out-of-sample performance even without explicit covariance regularisation. However, adding an upper bound (e.g., 5% or 10%) per firm would be a useful robustness check.

Source: Jagannathan, R. & Ma, T. (2003), "Risk Reduction in Large Portfolios: Why Imposing the Wrong Constraints Helps", Journal of Finance, 58(4), 1651-1684.

6. Ex-Post Performance Computation

6.1 Monthly Weight Drift

After the allocation is determined at end of year Y, the portfolio is held for 12 months without rebalancing. Each month, weights drift according to the relative price changes of the constituent securities:

alpha_{i,t+k} = alpha_{i,t+k-1} * (1 + R_{i,t+k}) / (1 + R_{p,t+k})

The monthly portfolio return is: **R_{p,t+k} = alpha'_{t+k-1} * R_{t+k}**

This buy-and-hold approach between annual rebalancing dates is standard and consistent with the project specification. Transaction costs are implicitly zero within each year.

Source: Project instructions, Section 2.2; Lecture 5, slides 35-36.

6.2 Performance Metrics

Metric	Formula / Method
Annualised arithmetic return	12 * mean(R_monthly) — Lecture 5, slide 12
Annualised geometric return	(1 + R_cumulative)^(12/T) - 1 — Lecture 5, slide 13
Annualised volatility	std(R_monthly) * sqrt(12) — Lecture 5, slide 12
Sharpe ratio	(mu_arith_ann - RF_ann) / sigma_ann — Lecture 5, slide 12
Maximum drawdown	max peak-to-trough loss from cumulative return series — Lecture 5, slide 10
Min / Max monthly return	Simple extrema of the monthly return series

6.3 Risk-Free Rate Treatment

The risk-free rate is sourced from the Fama-French research data file. Per explicit instructor guidance, the raw values (provided in percentage points, e.g., 0.41) are divided by 100 and treated as an annual rate, then compounded down to a monthly decimal: RF_monthly = (1 + RF_annual)^(1/12) - 1.

For the Sharpe ratio, the monthly RF is re-annualised as 12 * mean(RF_monthly) and subtracted from the annualised arithmetic return before dividing by annualised volatility.

Note: The standard Fama-French convention treats these values as monthly percentage returns. Our treatment as annual rates follows the instructor's explicit directive and produces systematically lower monthly RF values, which slightly inflates the Sharpe ratio. This should be acknowledged as a convention difference.

Source: Instructor directive (treat RF as annual rate); Fama, E.F. & French, K.R., Kenneth R. French Data Library.

6.4 Volatility Estimator Convention

The ex-post volatility uses the pandas .std() default, which divides by (T-1) (unbiased estimator), while the covariance matrix estimation uses the ML estimator (divides by T). This is a minor inconsistency. The project formula for the covariance matrix explicitly uses 1/T, while the lecture slides for ex-post volatility use 1/(T-1). We follow each specification in its respective context.

Source: Lecture 5, slides 6-7 (both estimators discussed); Project formula, Section 2.1 (ML for covariance).

7. Value-Weighted Benchmark

The value-weighted portfolio uses the same investment set as the minimum-variance portfolio (same eligible firms per year). Each month t, the weight of firm i is proportional to its market capitalisation at the end of month t-1:

w_{i,t} = Cap_{i,t-1} / sum_j(Cap_{j,t-1})

Firms with missing market cap at month t-1 receive zero weight (effectively excluded for that month). The same OOS delisting-aware return matrix (fill_oos_returns) is used for both the MV and VW portfolios, ensuring consistent treatment of delistings.

Aspect	Advantage	Disadvantage
Reweighting frequency	Monthly reweighting reflects actual market values (as an investor would hold)	Monthly turnover is higher than annual; however, no transaction costs are applied
Same universe	Direct comparability with the MV portfolio: same eligible firms, same return data	The VW benchmark is restricted to the EUR carbon-data universe, not a standard market index
Missing cap treatment	fillna(0) on market cap effectively drops firms with missing data for that month	A firm temporarily missing cap data gets zero weight, then may re-enter next month

Source: Project instructions, Section 2.3; Lecture 5, slide 38.

8. Summary: Design Choices vs Alternatives

Choice	Our Approach	Alternative	Why Ours	Flag
Cov estimation	Pairwise-complete + Ledoit-Wolf shrinkage	Listwise deletion or factor model	Maximises data; shrinkage guarantees PSD	None
Shrinkage target	Constant-correlation	Identity matrix (Ledoit-Wolf 2004b) or single-factor (Sharpe)	Preserves average dependence structure	Could compare targets
Stale threshold	30%	50% (project suggestion)	Stronger illiquidity filter	Report sensitivity
Delisting returns	-100% including OOS	-100% only in estimation window	Eliminates survivorship bias in OOS period	None
OOS missing returns	Delisting-aware fill (fill_oos_returns)	Blind fillna(0)	Distinguishes suspension from delisting	None
Optimiser	SLSQP with analytical Jacobian	cvxpy / quadprog	Sufficient for QP with inequality constraints	None

Weight bounds	[0, None]	[0, 0.05] or [0, 0.10]	Matches project specification	Discuss concentration
RF treatment	Annual rate compounded to monthly	Monthly return / 100	Instructor directive	Convention differs from standard FF
Vol estimator	1/(T-1) ex-post, 1/T for covariance	Consistent 1/T everywhere	Follows each source formula	Minor inconsistency

9. References

Chen, Y., Wiesel, A., Eldar, Y. & Hero, A. (2010). Shrinkage Algorithms for MMSE Covariance Estimation. IEEE Transactions on Signal Processing, 58(10), 5016-5029.

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